

1) Why does an over-determined system usually not have an exact solution?

When the data includes some bit of error, and we have as many equations as unknowns, we can find an exact solution, even though there will be error in that solution, it will be the best fit.

If on the other hand we have more equations than unknowns, we will encounter contradictions, making an exact solution impossible, unless the data happens to perfectly line up — which is extremely unlikely.

2) In one sentence, what operation is least squares performing?

Least squares is performing a linear regression, which projects the data onto the subspace of predictions in such a way as to minimize the error between the prediction and the data.

3) Suppose the best approximation is $A\hat{x}$. What is true of the residual vector $\vec{e} = b - A\hat{x}$?

The residual vector is orthogonal to every vector in the column space, and it's the vector

that minimizes the error in the approximation.
It has the same dimension of $\vec{b}$.

4) Given $A_{10 \times 3}$, what are the dimensions of

- $A^T - 3 \times 10$
- $A^T A \quad 3 \times 10 \quad 10 \times 3 = 3 \times 3$
- $A^T \vec{b} \quad 3 \times 10 \quad 10 \times 1 = 3 \times 1$

5) Starting with $A^T(\vec{b} - A\vec{x}) = 0$, expand it algebraically to obtain the normal equations

$$A^T(\vec{b} - A\vec{x}) = 0$$

$$A^T \vec{b} - A^T A \vec{x} = 0$$

$$A^T \vec{b} = A^T A \vec{x}$$

$$(A^T A)^{-1} A^T \vec{b} = \vec{x}$$

Dimensional check

$$(2 \times 4 \quad 4 \times 2) \quad 2 \times 4 \quad 4 \times 1$$

$$\underbrace{2 \times 2} \quad \underbrace{2 \times 4} \quad 4 \times 1 \rightarrow 2 \times 1$$